Netflix Global Content Analysis and Visualization using Python (Pandas) and Power BI

1. Objective

The objective of this project is to perform a comprehensive analysis of Netflix's content catalogue using Python (Pandas) and Power BI. The project aims to transform raw data into meaningful business insights through data cleaning, preprocessing, transformation, exploratory analysis, and interactive dashboard visualization.

The project focuses on understanding content distribution across different countries, genres, ratings, and release years while identifying trends in Netflix's content growth over time. The final outcome is an interactive Power BI dashboard that enables users to explore the dataset and derive actionable insights.

Specific objectives include:

• Cleaning and preparing the raw dataset for analysis.

• Analysing the distribution of Movies and TV Shows.

• Identifying trends in content additions over the years.

• Evaluating country-wise content production.

• Examining genre and rating distributions.

• Building interactive visualizations using Power BI.

• Generating business recommendations based on analytical findings.

2. Dataset Description

The dataset used in this project is the Netflix Titles Dataset, which contains information about movies and television shows available on Netflix.

The dataset provides metadata related to content type, title, cast, director, country, release year, ratings, duration, and genres. It serves as an excellent source for understanding Netflix's content strategy and market distribution.

Dataset Characteristics:

• Dataset Name: Netflix_Database.csv

• File Format: CSV

• Number of Records: 8,804

• Number of Columns: 12

• Source: Netflix Content Dataset

• Tools Used: Google Colab, Pandas, Power BI

Netflix Global Content Analysis and Visualization using Python (Pandas) and Power BI

The dataset initially contained missing values and inconsistencies which required preprocessing before analysis. After cleaning and transformation, the resulting dataset was used for exploratory analysis and dashboard development.

3. Table Explanation

The dataset consists of the following attributes:

show_id:
Unique identifier assigned to each Netflix title.

type:
Specifies whether the content is a Movie or TV Show.

title:
Name of the movie or television show.

director:
Director associated with the content.

cast:
List of actors and actresses featured in the content.

country:
Country where the content was produced.

date_added:
Date when the content was added to Netflix.

release_year:
Original release year of the content.

rating:
Audience maturity rating assigned to the content.

duration:
Movie runtime in minutes or number of seasons for TV shows.

listed_in:
Genre or category assigned to the content.

description:
Brief synopsis of the content.

4. MECE Breakdown

The dataset was analysed using the MECE (Mutually Exclusive, Collectively Exhaustive) framework to ensure complete and non-overlapping categorization of information.

A. Content Information

Netflix Global Content Analysis and Visualization using Python (Pandas) and Power BI

• Show ID

• Title

• Type

• Description

B. Production Information

• Director

• Cast

• Country

C. Time Information

• Release Year

• Date Added

D. Classification Information

• Rating

• Genre

E. Duration Information

• Duration

Based on this categorization, the analysis addresses important business questions related to content distribution, audience targeting, geographical presence, and content growth trends.

5. Data Connection

The project utilized two major tools: Google Colab for data processing and Power BI for visualization.

Google Colab Connection Process:

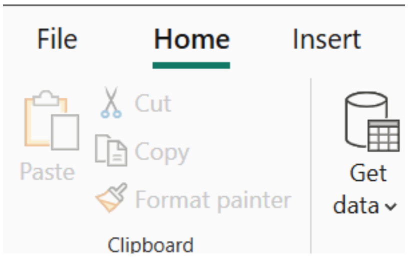
The raw Netflix CSV dataset was stored in Google Drive and imported into Google Colab by mounting the drive using the drive.mount() function from the google. Colab library. After establishing the connection, the dataset was loaded into a Pandas DataFrame using the pd.read_csv() function by specifying the file path of the CSV file. The resulting DataFrame was named df and served as the primary dataset for subsequent cleaning, transformation, and analysis tasks.

```
#Importing Libraries:
import pandas as pd
from google.colab import drive
#Load Dataset:
drive.mount('/content/drive')
df=pd.read_csv('/content/drive/MyDrive/Netflix_Database.csv')
```

Netflix Global Content Analysis and Visualization using Python (Pandas) and Power BI

Power BI connection process:

After the raw data had been cleaned, processed, and transformed using Pandas, the finalized dataset was imported into Power BI Desktop through the Get Data functionality. The imported data was then used to perform further analysis, create visualizations, and develop an interactive dashboard for presenting key insights and findings derived from the Netflix dataset.



6. Data cleaning, processing and transformation:

Before the dataset was used for analysis, several preliminary inspections were conducted to assess its quality and structure. These inspections included identifying missing (null) values, checking for datatype inconsistencies, reviewing column names for standardization, and examining the dataset for any other data quality issues. Based on the findings, the necessary cleaning and transformation steps were performed to ensure the dataset was accurate, consistent, and analysis-ready.

```
#Basic Inspection:
df.info()
df.shape
df.isnull().sum()
```

Following the inspection, the necessary data transformations were performed to improve data quality and usability. These transformations included renaming column headers to make them more meaningful and easier to understand, converting date-related columns to the appropriate date format, and standardizing the dataset structure for analysis. After the transformation process was completed, the cleaned DataFrame was exported to Google Drive in Excel format for further analysis and visualization.

```
#Standardize Column Names:
df.columns = df.columns.str.lower().str.replace(" ","_")
#Convert Date Column:
df['date_added'] = pd.to_datetime(df['date_added'])
df['year_added'] = df['date_added'].dt.year
#Save Cleaned Dataset:
df.to_excel("/content/drive/MyDrive/Netflix_df.xlsx",index=False)
```

To facilitate more effective analysis of country-wise and genre-wise trends, two additional DataFrames, namely country_df and genre_df, were created by copying the original dataset. In the original dataset, the country and listed_in columns contained multiple values within a single row, separated by commas. Therefore, these columns were first split into individual values and then expanded using the explode operation. This transformation ensured that each country and genre appeared in a separate row, making aggregation, filtering, and trend analysis significantly easier and more accurate. The resulting DataFrames were subsequently exported to Google Drive for further analysis and visualization.

```
#Creating country dataset by exploding country column:
country_df = df.copy()
country_df['country'] = country_df['country'].str.split(',')
country_df = country_df.explode('country')
country_df['country'] = country_df['country'].str.strip()
```

```
#Splitting and exploding genre:
genre_df = df.copy()
genre_df['listed_in'] = genre_df['listed_in'].str.split(',')
genre_df = genre_df.explode('listed_in')
genre_df['listed_in'] = genre_df['listed_in'].str.strip()
```

```
#Saving the country df:
country_df.to_excel("/content/drive/MyDrive/country_df.xlsx",index=False)
```

```
#Saving genre dfs:
genre_df.to_excel("/content/drive/MyDrive/genre_df.xlsx",index=False)
```

7. Objectives and Derived Insights:

Country-wise Netflix Content Distribution

To analyse the distribution of Netflix content across different countries, the country_df DataFrame was grouped by the country column using the groupby() function. The unique count of show_id values was then computed for each country to determine the total number of distinct titles available. Finally, the aggregated results were sorted in descending order of title count, enabling the identification of the top content-producing countries on Netflix.

```
#Creating country dataset by exploding country column:
country_df = df.copy()
country_df['country'] = country_df['country'].str.split(',')
country_df = country_df.explode('country')
country_df['country'] = country_df['country'].str.strip()
top_countries = country_df.groupby('country')['show_id']\
                          .nunique()\
                          .sort_values(ascending=False)\
                          .reset_index()
print(top_countries.head(20))
```

Netflix Global Content Analysis and Visualization using Python (Pandas) and Power BI

	country	show_id
0	United States	3687
1	India	1046
2	Unknown Country	831
3	United Kingdom	806
4	Canada	445
5	France	393
6	Japan	318
7	Spain	232
8	South Korea	231
9	Germany	226
10	Mexico	169
11	China	162
12	Australia	160
13	Egypt	117
14	Turkey	113
15	Hong Kong	105
16	Nigeria	103
17	Italy	100
18	Brazil	97
19	Argentina	91

Insights from Country-wise Netflix Content Distribution Analysis

- The United States is the largest contributor to Netflix's content library, followed by India and the United Kingdom, making them the top content-producing countries on the platform.
- India is the leading contributor in Asia, reflecting Netflix's growing investment in regional and South Asian content.
- A significant number of titles fall under the "Unknown Country" category, indicating potential gaps in country-related metadata.
- Countries such as Canada, France, Japan, Germany, and Australia also contribute substantially, highlighting Netflix's partnerships with major entertainment industries.
- Several Asian countries, including Japan, South Korea, China, Hong Kong, and Turkey, rank among the top contributors, showcasing Netflix's expanding focus on Asian markets.
- The content distribution follows a long-tail pattern, where a few countries contribute most titles while many others contribute smaller amounts, resulting in a globally diverse but concentrated content library.

Genre Popularity Analysis

To derive genre-based insights, the previously created genre_df DataFrame was utilized. Since the listed_in column contained multiple genres within a single record, the column was first split and exploded so that each genre appeared in a separate row. A groupby() operation was then performed on the listed_in column, and the count of show_id values was calculated for each genre to determine

Netflix Global Content Analysis and Visualization using Python (Pandas) and Power BI

genre popularity across the Netflix catalogue. The resulting dataset was subsequently sorted in descending order based on the title count to identify the most prevalent genres.

```
#Splitting and exploding genre:
genre_df = df.copy()
genre_df['listed_in'] = genre_df['listed_in'].str.split(',')
genre_df = genre_df.explode('listed_in')
genre_df['listed_in'] = genre_df['listed_in'].str.strip()
#Genre Counts:
genre_count = genre_df.groupby('listed_in')['show_id']\
                        .count()\
                        .sort_values(ascending=False)\
                        .reset_index()
print(genre_count)
```

```
                   listed_in  show_id
0       International Movies     2752
1                     Dramas     2427
2                   Comedies     1674
3     International TV Shows     1351
4              Documentaries      869
5         Action & Adventure      859
6                  TV Dramas      763
7         Independent Movies      756
8   Children & Family Movies      641
9            Romantic Movies      616
10               TV Comedies      581
11                 Thrillers      577
12            Crime TV Shows      470
13                 Kids' TV       451
14                Docuseries      395
15          Music & Musicals      375
16         Romantic TV Shows      370
17             Horror Movies      357
18           Stand-Up Comedy      343
19                Reality TV      255
20          British TV Shows      253
21           Sci-Fi & Fantasy      243
22             Sports Movies      219
23              Anime Series      176
24   Spanish-Language TV Shows    174
```

```
25           TV Action & Adventure      168
26                Korean TV Shows       151
27                 Classic Movies       116
28                   LGBTQ Movies       102
29                   TV Mysteries        98
30              Science & Nature TV      92
31             TV Sci-Fi & Fantasy       84
32                      TV Horror        75
33                 Anime Features        71
34                    Cult Movies        71
35                  Teen TV Shows        69
36            Faith & Spirituality       65
37                    TV Thrillers       57
38   Stand-Up Comedy & Talk Shows        56
39                         Movies        54
40               Classic & Cult TV       28
41                       TV Shows        16
```

Additionally, a multi-column groupby() operation was performed on both the listed_in and type columns to analyse the distribution of Movies and TV Shows across different genres. This enabled a more detailed understanding of genre preferences within each content type and helped identify the dominant genres for Movies and TV Shows separately.

```
#Genre by Content Type:
genre_type = genre_df.groupby(
    ['listed_in','type']
)['show_id'].count().reset_index()
print(genre_type)
```

```
                    listed_in      type  show_id
0          Action & Adventure     Movie      859
1              Anime Features     Movie       71
2                Anime Series   TV Show      176
3             British TV Shows  TV Show      253
4      Children & Family Movies     Movie      641
5            Classic & Cult TV  TV Show       28
6              Classic Movies     Movie      116
7                    Comedies     Movie     1674
8              Crime TV Shows   TV Show      470
9                  Cult Movies     Movie       71
10              Documentaries     Movie      869
11                 Docuseries   TV Show      395
12                     Dramas     Movie     2427
13       Faith & Spirituality     Movie       65
14              Horror Movies     Movie      357
15         Independent Movies     Movie      756
16      International Movies     Movie     2752
17    International TV Shows   TV Show     1351
18                   Kids' TV   TV Show      451
19             Korean TV Shows  TV Show      151
20               LGBTQ Movies     Movie      102
21                     Movies     Movie       54
22           Music & Musicals     Movie      375
23                 Reality TV   TV Show      255
24            Romantic Movies     Movie      616
25          Romantic TV Shows   TV Show      370
26           Sci-Fi & Fantasy     Movie      243
27        Science & Nature TV   TV Show       92
```

Netflix Global Content Analysis and Visualization using Python (Pandas) and Power BI

```
28     Spanish-Language TV Shows   TV Show    174
29                 Sports Movies     Movie    219
30               Stand-Up Comedy     Movie    343
31  Stand-Up Comedy & Talk Shows   TV Show     56
32          TV Action & Adventure   TV Show    168
33                   TV Comedies   TV Show    581
34                     TV Dramas   TV Show    763
35                     TV Horror   TV Show     75
36                  TV Mysteries   TV Show     98
37            TV Sci-Fi & Fantasy   TV Show     84
38                      TV Shows   TV Show     16
39                  TV Thrillers   TV Show     57
40                 Teen TV Shows   TV Show     69
41                     Thrillers     Movie    577
```

Summary of Genre Popularity Analysis

- International Movies, Dramas, and Comedies are the most popular genres on Netflix, indicating strong audience demand for diverse and entertainment-focused content.
- International TV Shows also have a significant presence, highlighting Netflix's focus on expanding its global content library.
- Movies dominate the platform across most genres, suggesting that Netflix's catalogue is primarily movie-oriented.
- Documentary, Action & Adventure, and Independent Movie categories contribute substantially to the overall content distribution.
- The presence of genres such as Anime Series, Korean TV Shows, and Docuseries reflects Netflix's investment in niche and internationally trending content.
- Overall, Netflix maintains a highly diversified content portfolio that caters to audiences with varying preferences across different regions and age groups.

Audience Segmentation using Ratings

To analyse audience segmentation based on content ratings, the cleaned Netflix DataFrame (df) was utilized. A groupby() operation was performed on the rating column, and the count of show_id values was calculated for each rating category to determine the distribution of content across different audience segments. The resulting DataFrame was then sorted in descending order based on the title count to identify the most common content ratings on Netflix.

```
#Rating Distribution:
rating_count = df.groupby('rating')['show_id']\
                .count()\
                .sort_values(ascending=False)\
                .reset_index()
print(rating_count)
```

Netflix Global Content Analysis and Visualization using Python (Pandas) and Power BI

```
        rating  show_id
0        TV-MA     3207
1        TV-14     2160
2        TV-PG      863
3            R      799
4        PG-13      490
5        TV-Y7      334
6         TV-Y      307
7           PG      287
8         TV-G      220
9           NR       80
10           G       41
11    TV-Y7-FV        6
12   Not Rated        4
13       NC-17        3
14          UR        3
```

Additionally, a multi-column groupby() operation was performed on the rating and type columns. The count of show_id values was calculated for each rating-category and content-type combination (Movie or TV Show). This analysis provided deeper insights into audience preferences and helped identify how Netflix distributes Movies and TV Shows across different age and maturity rating categories.

```
#Rating by type:
rating_type = df.groupby(
    ['rating','type']
)['show_id'].count().reset_index()
print(rating_type)
```

```
        rating     type  show_id
0            G    Movie       41
1        NC-17    Movie        3
2           NR    Movie       75
3           NR  TV Show        5
4    Not Rated    Movie        2
5    Not Rated  TV Show        2
6           PG    Movie      287
7        PG-13    Movie      490
8            R    Movie      797
9            R  TV Show        2
10       TV-14    Movie     1427
11       TV-14  TV Show      733
12        TV-G    Movie      126
```

Netflix Global Content Analysis and Visualization using Python (Pandas) and Power BI

```
13        TV-G  TV Show     94
14       TV-MA    Movie   2062
15       TV-MA  TV Show   1145
16       TV-PG    Movie    540
17       TV-PG  TV Show    323
18        TV-Y    Movie    131
19        TV-Y  TV Show    176
20       TV-Y7    Movie    139
21       TV-Y7  TV Show    195
22    TV-Y7-FV    Movie      5
23    TV-Y7-FV  TV Show      1
24          UR    Movie      3
```

Summary of Audience Segmentation Analysis

- TV-MA is the most common rating category on Netflix, followed by TV-14 and TV-PG, indicating a strong focus on content targeted toward mature and teenage audiences.
- The majority of TV-MA, TV-14, and TV-PG titles are Movies, although these ratings also have substantial representation among TV Shows.
- Family-friendly categories such as TV-Y, TV-Y7, TV-G, and G contain significantly fewer titles compared to mature audience ratings.
- R-rated and PG-13 content have a strong presence in the Netflix catalogue, reflecting continued investment in content for adult viewers.
- The rating distribution suggests that Netflix primarily caters to teenagers and adults while maintaining a smaller but notable collection of children's and family-oriented content.
- Overall, Netflix adopts a balanced content strategy by serving multiple audience segments, with a stronger emphasis on mature and young-adult viewers.

Content Growth Analysis

To analyse Netflix's content growth over time, the cleaned DataFrame (df) was utilized. A groupby() operation was performed on the year_added column, and the count of show_id values was calculated for each year to determine the number of titles added to Netflix annually. The resulting DataFrame was then used to identify overall content growth trends across different years.

```
#Year-Wise content addition:
growth_df = df.groupby('year_added')['show_id']\
              .count()\
              .reset_index()
growth_df.rename(
    columns={'show_id':'total_titles'},
    inplace=True)
print(growth_df)
```

Netflix Global Content Analysis and Visualization using Python (Pandas) and Power BI

```
    year_added  total_titles
0         1900            10
1         2008             2
2         2009             2
3         2010             1
4         2011            13
5         2012             3
6         2013            11
7         2014            24
8         2015            82
9         2016           427
10        2017          1187
11        2018          1649
12        2019          2016
13        2020          1879
14        2021          1498
```

Additionally, a multi-column groupby() operation was performed on the year_added and type columns. The count of show_id values was calculated for each combination of year and content type (Movie or TV Show). This analysis helped determine how the addition of Movies and TV Shows evolved over time and provided insights into Netflix's content acquisition and expansion strategy. The aggregated results were subsequently sorted based on title counts to highlight the years with the highest content additions.

```
#Type-wise Growth:
growth_type = df.groupby(
    ['year_added','type']
)['show_id'].count().reset_index()
print(growth_type)
```

```
    year_added     type  show_id
0         1900  TV Show       10
1         2008    Movie        1
2         2008  TV Show        1
3         2009    Movie        2
4         2010    Movie        1
5         2011    Movie       13
6         2012    Movie        3
7         2013    Movie        6
8         2013  TV Show        5
9         2014    Movie       19
10        2014  TV Show        5
11        2015    Movie       56
12        2015  TV Show       26
13        2016    Movie      251
14        2016  TV Show      176
15        2017    Movie      838
16        2017  TV Show      349
17        2018    Movie     1237
18        2018  TV Show      412
19        2019    Movie     1424
20        2019  TV Show      592
21        2020    Movie     1284
22        2020  TV Show      595
23        2021    Movie      993
24        2021  TV Show      505
```

Netflix Global Content Analysis and Visualization using Python (Pandas) and Power BI

Summary of Content Growth Analysis

- Netflix experienced relatively slow content growth before 2015, with fewer than 100 titles being added annually in most years.
- A significant surge in content additions began in 2016, indicating a major expansion phase in Netflix's content acquisition strategy.
- The platform witnessed rapid growth between 2017 and 2020, with annual additions increasing from 1,187 titles in 2017 to a peak of 2,016 titles in 2019.
- Although content additions remained high in 2020 (1,879 titles), a slight decline was observed in 2021, with 1,498 titles added.
- Movies consistently outnumbered TV Shows across all years, making them the primary driver of Netflix's content growth.
- The overall trend demonstrates Netflix's aggressive global expansion strategy and substantial investment in content acquisition during the late 2010s.

Key Performance Index

To obtain a high-level overview of the Netflix dataset, a Key Performance Indicator (KPI) dataset was created using Pandas. The KPI dataset was designed to summarize the overall size and diversity of Netflix's content library through key metrics such as Total Titles, Total Movies, Total TV Shows, Total Countries, and Total Genres.

The KPI values were calculated using various Pandas functions and then stored as key-value pairs in a separate DataFrame named kpi. The Total Titles metric was calculated by counting the unique show_id values in the main DataFrame (df). Total Movies and Total TV Shows were determined by filtering the type column for "Movie" and "TV Show" respectively and counting the resulting records. The Total Countries metric was calculated by counting the distinct country values in the country_df DataFrame, while the Total Genres metric was obtained by counting the unique genre values in the genre_df DataFrame.

The resulting KPI dataset provides a concise summary of Netflix's content catalogue and serves as the foundation for dashboard-level performance indicators in Power BI.

```python
#KPI Dataset:
kpi = pd.DataFrame({
    'Metric':[
        'Total Titles',
        'Movies',
        'TV Shows',
        'Countries',
        'Genres'
    ],
    'Value':[
        df['show_id'].nunique(),
        df[df['type']=='Movie'].shape[0],
        df[df['type']=='TV Show'].shape[0],
        country_df['country'].nunique(),
        genre_df['listed_in'].nunique()
    ]
})

print(kpi)
```

Netflix Global Content Analysis and Visualization using Python (Pandas) and Power BI

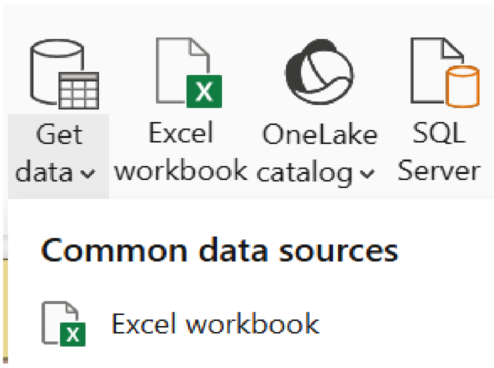
	Metric	Value
0	Total Titles	8804
1	Movies	6128
2	TV Shows	2676
3	Countries	124
4	Genres	42

Summary of KPI Analysis

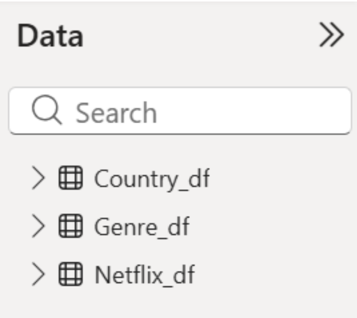
- Netflix's content library contains 8,804 unique titles, indicating a vast and diverse collection of entertainment content.
- Movies dominate the platform with 6,128 titles, accounting for approximately 70% of the total catalogue.
- TV Shows contribute 2,676 titles, demonstrating Netflix's substantial investment in episodic content.
- The content originates from 124 different countries, highlighting Netflix's strong global presence and international reach.
- The platform offers content across 42 distinct genres, showcasing a highly diversified portfolio designed to cater to a wide range of audience preferences.
- The KPI metrics collectively indicate that Netflix maintains a large, globally distributed, and genre-diverse content library, supporting its position as a leading streaming platform.

8. Power BI dashboard development

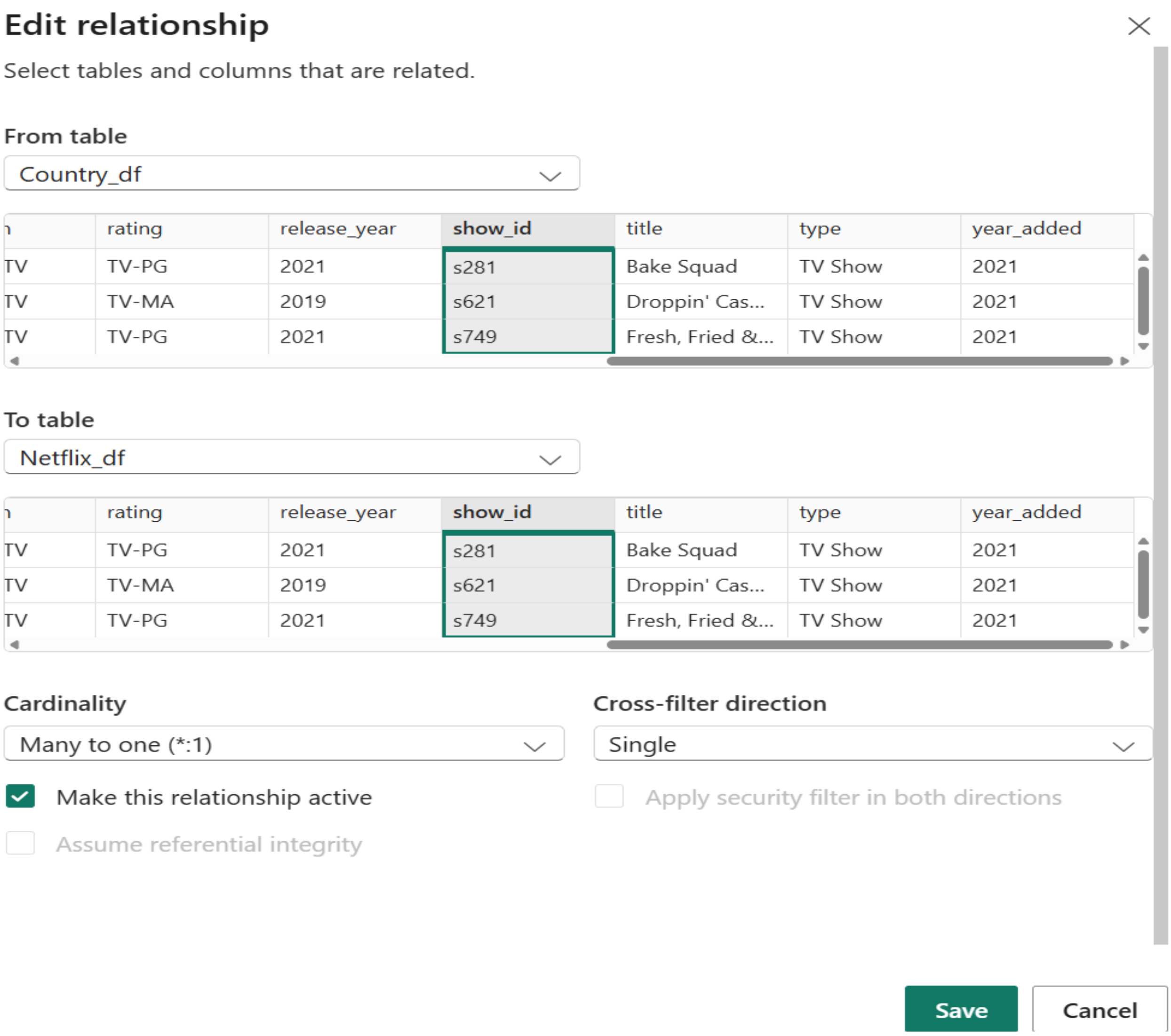
Before developing the Power BI dashboard, the three datasets generated during the Pandas analysis phase—Netflix_df (cleaned Netflix dataset), country_df, and genre_df—were exported to Google Drive in Excel format and subsequently imported into Power BI Desktop using the Get Data → Excel Workbook functionality.



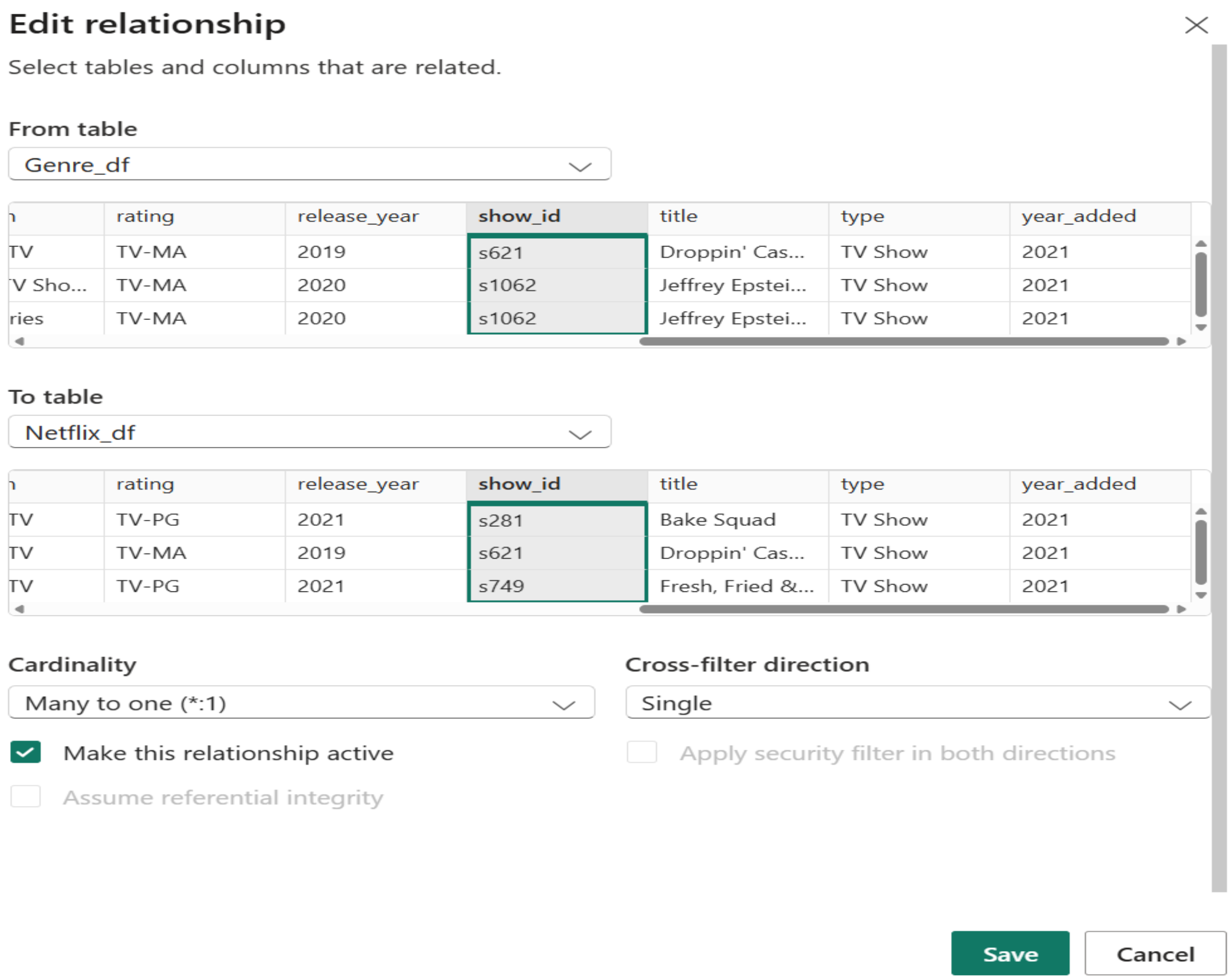
Netflix Global Content Analysis and Visualization using Python (Pandas) and Power BI



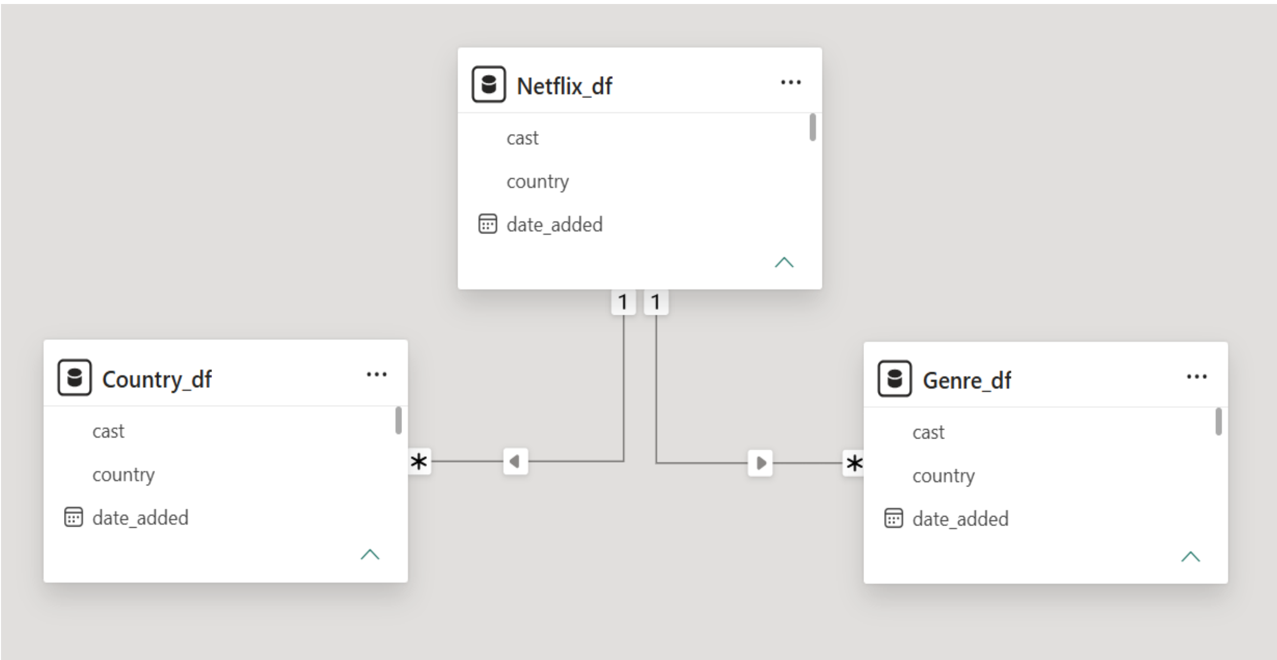
After loading the datasets, the required data model was established in the Model View. One-to-many relationships were created between Netflix_df and country_df, as well as between Netflix_df and genre_df, using the show_id column as the common key. To prevent ambiguous filter propagation and ensure accurate analysis, the cross-filter direction was configured appropriately by disabling bidirectional filtering. This data modelling approach enabled efficient interaction among the datasets while maintaining data integrity and optimal dashboard performance.



Netflix Global Content Analysis and Visualization using Python (Pandas) and Power BI



The established ER Diagram is shown below:



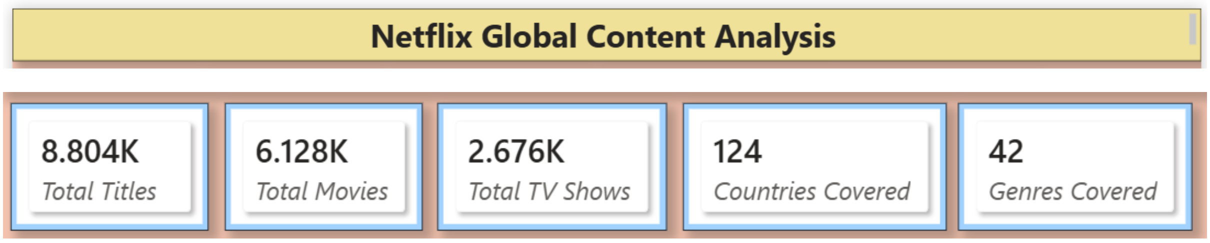
Netflix Global Content Analysis and Visualization using Python (Pandas) and Power BI

To effectively present the analytical findings, a multi-page Power BI dashboard was developed consisting of five dedicated report pages.

Page 1: Netflix Overview

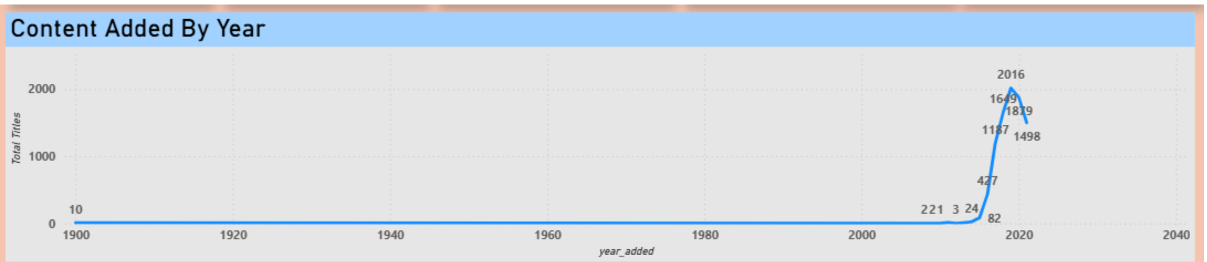
This page serves as the main dashboard overview. It includes the report title, key performance indicator (KPI) cards such as Total Titles, Total Movies, Total TV Shows, Total Countries, and Total Genres. Additionally, visualizations comparing Movies vs TV Shows and Year-wise Content Growth were included. Interactive slicers such as Type, Year Added, Rating, Country, and Genre were also provided to enable dynamic filtering.

The dashboard development process began by inserting a text box on the report page and adding the title "Netflix Global Content Analysis". Subsequently, five measures were created to calculate key performance indicators (KPIs), namely Total Titles, Total Movies, Total TV Shows, Total Genres, and Total Countries. These measures were displayed using KPI card visuals, and the callout values were formatted as whole numbers to ensure accurate and readable data representation. All KPI cards were then arranged horizontally at the top of the dashboard to provide a concise overview of the Netflix content library.



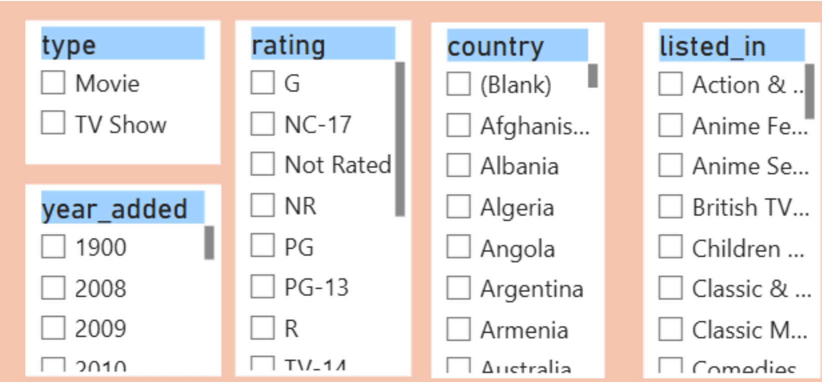

```
Total Titles = DISTINCTCOUNT(Netflix_df[show_id])

Total Movies = CALCULATE(COUNT(Netflix_df[show_id]),Netflix_df[type]="Movie")

Total TV Shows = CALCULATE(COUNT(Netflix_df[show_id]),Netflix_df[type]="TV Show")

Countries Covered = DISTINCTCOUNT(Country_df[country])

Genres Covered = DISTINCTCOUNT(Genre_df[listed_in])
```

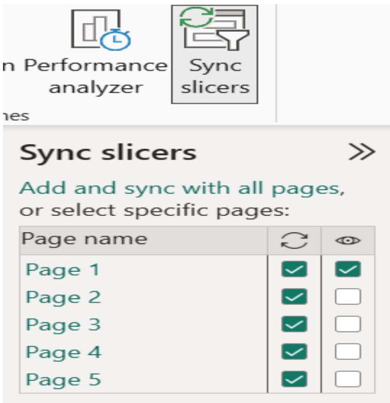
Next, a line chart was added to visualize Netflix's content growth over time. The year_added field was placed on the X-axis, while the Total Titles measure was placed on the Y-axis. Data labels were enabled to display the exact number of titles added each year, improving the interpretability of the visualization.



Netflix Global Content Analysis and Visualization using Python (Pandas) and Power BI

Finally, interactive slicers were added for key dimensions such as Content Type, Rating, Country, and Genre (listed_in). To ensure a consistent filtering experience across all report pages, the slicers were synchronized using the View → Sync Slicers functionality. This enabled users to apply a filter once and have it reflected throughout the entire dashboard.





To enhance user experience and provide a structured summary of the analysis, four navigation buttons were added to the Netflix Overview page. These buttons allow users to seamlessly navigate to the Country Overview, Genre Overview, Audience Segmentation Overview, and Content Growth Overview pages. By enabling quick access to different sections of the dashboard, the navigation buttons improve interactivity, usability, and overall user engagement, making the dashboard more intuitive and user-friendly.

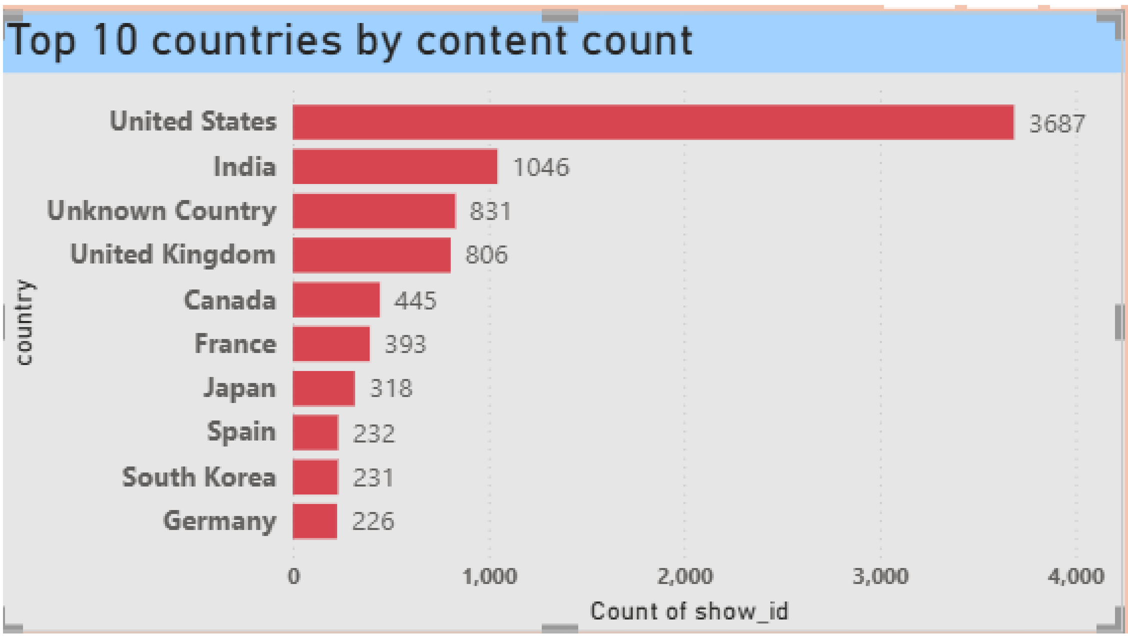
Netflix Global Content Analysis and Visualization using Python (Pandas) and Power BI

Page 2: Country Overview

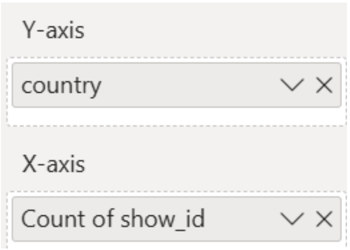
This page focuses on the geographical distribution of Netflix content. It presents country-wise content availability and analyzes the distribution of Movies and TV Shows across different countries.

In this page, the first visualization created was a Map Visual to illustrate the country-wise distribution of Netflix content across the globe. The country field was placed in the Location section to plot the content geographically. To enhance the visualization and better represent content volume, the unique count of show_id was added to the Bubble Size section. As a result, the size of each bubble corresponds to the number of Netflix titles available from a particular country, allowing users to easily identify regions with high and low content contributions. This visualization provides a clear overview of Netflix's global content presence and geographical reach.

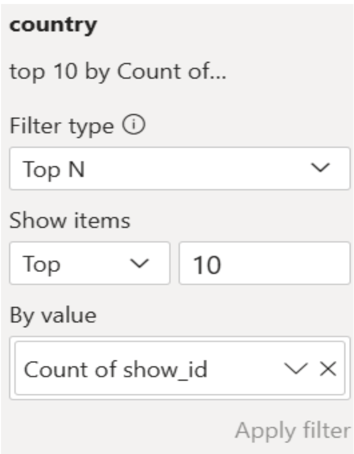
Below the map visualization, a Clustered Bar Chart was added to highlight the Top 10 Countries by Content Count. In this visual, the unique count of show_id was placed on the X-axis, while the country field from the country_df table was placed on the Y-axis. Data labels were enabled to display the exact number of titles contributed by each country, improving the readability and interpretability of the chart.



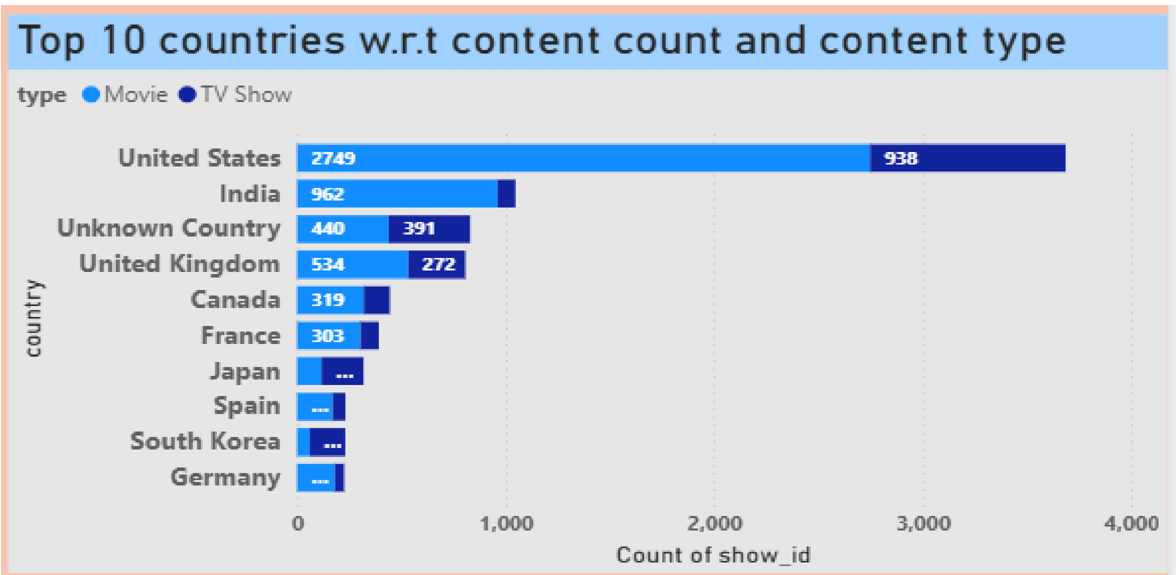
Netflix Global Content Analysis and Visualization using Python (Pandas) and Power BI



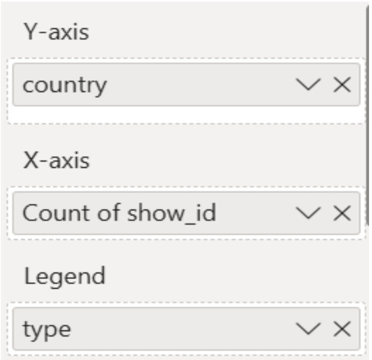
To focus on the leading content-producing countries, a Top N filter was applied to the country field. The value of N was set to 10, and the ranking was based on the unique count of show_id. This allowed the visualization to display only the top 10 countries with the highest number of Netflix titles, providing a clear comparison of their contributions to the platform's content library.



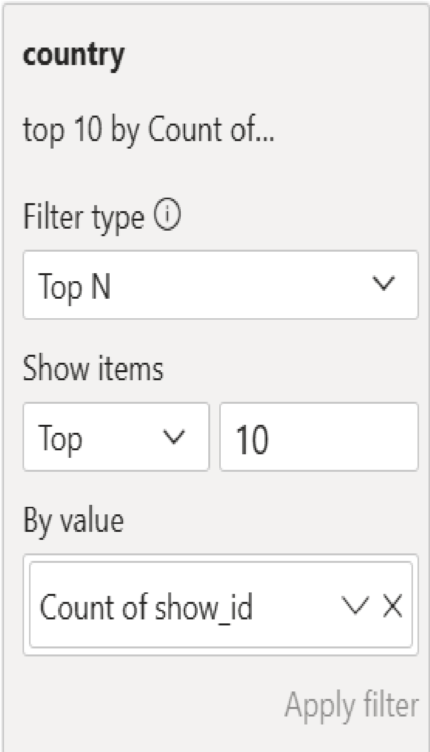
After the second visualization, a Stacked Bar Chart was created to analyse the Top 10 Countries by Content Count and Content Type. In this visual, the unique count of show_id was placed on the X-axis, while the country field from the country_df table was placed on the Y-axis. The type column was added to the Legend section to differentiate the content into Movies and TV Shows, enabling a comparative analysis of content types across countries.



Netflix Global Content Analysis and Visualization using Python (Pandas) and Power BI



To focus on the most significant contributors, a Top N filter was applied to the country field. The value of N was set to 10, and the ranking was based on the unique count of show_id. This filtering ensured that only the top 10 countries with the highest number of Netflix titles were displayed. The visualization provides a clear comparison of how Movies and TV Shows contribute to the overall content library within each of the leading content-producing countries.

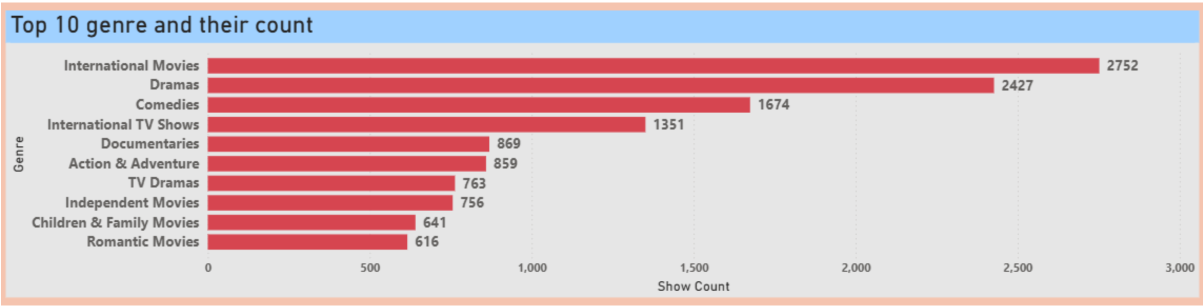


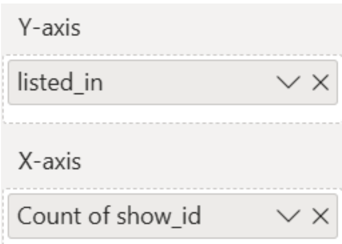
Page 3: Genre Overview

This page provides insights into genre-based content distribution. It highlights the most popular genres on Netflix and compares the number of Movies and TV Shows within each genre category.

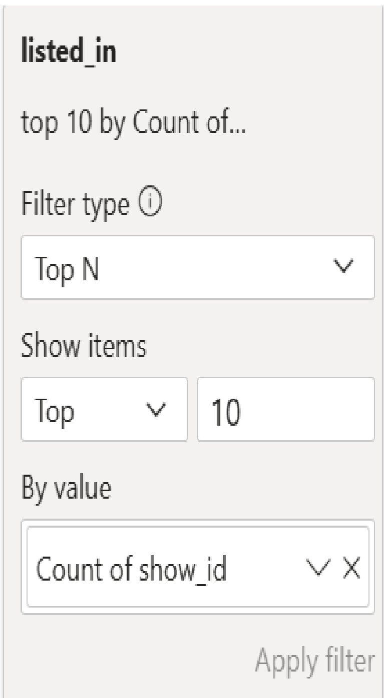
In this page, the first visualization created was a Clustered Bar Chart to showcase the Top 10 Netflix Genres by Title Count. For this visual, the unique count of show_id was placed on the X-axis, while the listed_in (genre) column from the genre_df table was placed on the Y-axis. Data labels were enabled to display the exact number of titles within each genre, enhancing the clarity and readability of the visualization.

Netflix Global Content Analysis and Visualization using Python (Pandas) and Power BI



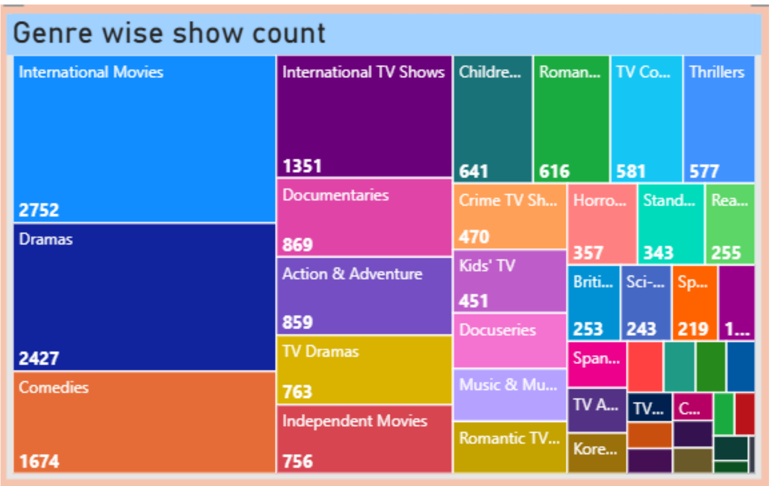


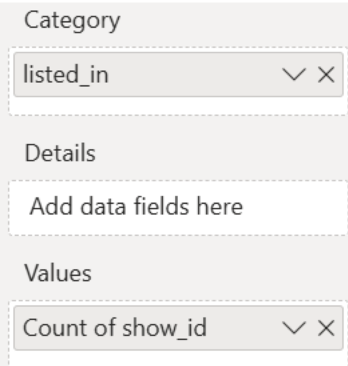
To focus on the most prominent genres, a Top N filter was applied to the listed_in column. The value of N was set to 10, and the ranking was based on the unique count of show_id. This filtering ensured that only the top 10 genres with the highest number of Netflix titles were displayed, allowing for a clear comparison of the platform's most popular content categories.



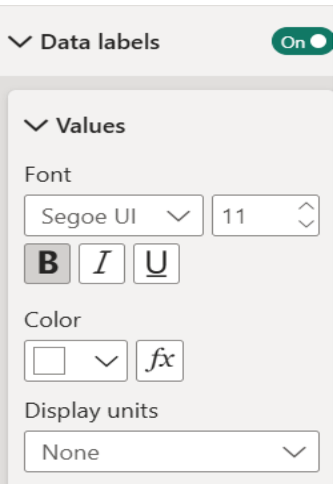
As the second visualization on this page, a Treemap was created to illustrate the distribution of Netflix titles across different genres. In this visual, the listed_in (genre) column from the genre_df table was placed in the Category section, while the count of show_id was placed in the Values section. This allowed each genre to be represented by a rectangle whose size corresponds to the number of titles within that genre.

Netflix Global Content Analysis and Visualization using Python (Pandas) and Power BI



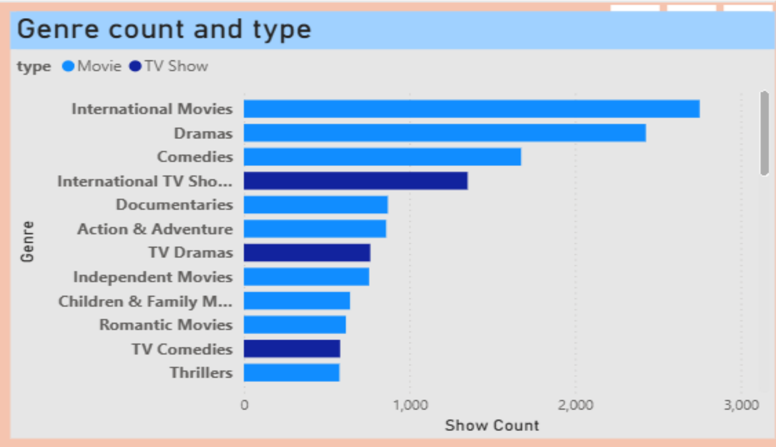


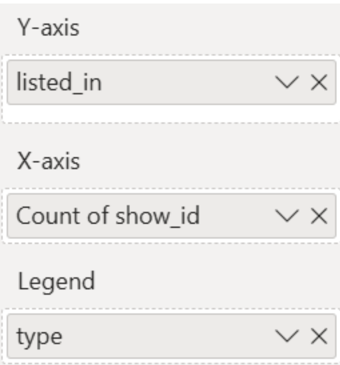
To enhance readability and provide precise information, data labels were enabled and formatted as whole numbers. The Treemap offers an intuitive visual representation of genre-wise content distribution, making it easy to identify the most and least represented genres within Netflix's content library.



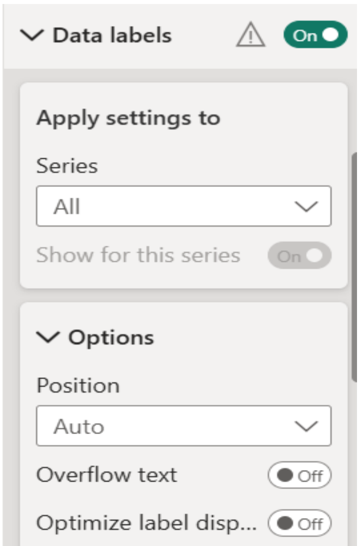
As the third visualization on this page, a Stacked Bar Chart was created to analyse genre distribution by content type. In this visual, the unique count of show_id was placed on the X-axis, while the listed_in (genre) column from the genre_df table was placed on the Y-axis. The type column was added to the Legend section to differentiate the content into Movies and TV Shows, enabling a comparative analysis of content types within each genre.

Netflix Global Content Analysis and Visualization using Python (Pandas) and Power BI





To improve readability and provide precise information, data labels were enabled and formatted as whole numbers. This visualization helps identify the contribution of Movies and TV Shows across different genres and provides a deeper understanding of Netflix's genre-wise content composition.

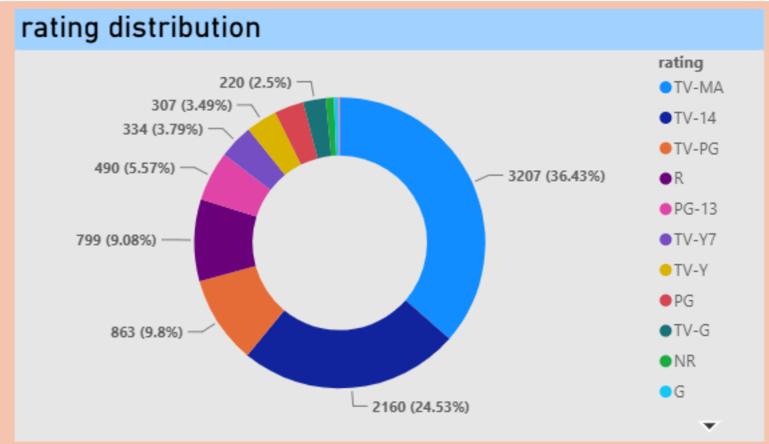


Page 4: Audience Segmentation Overview

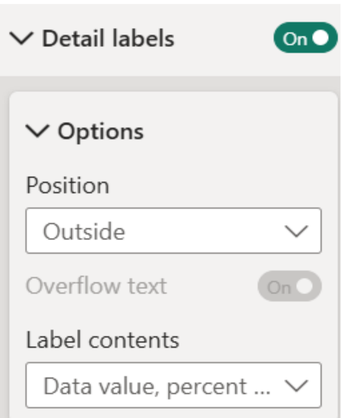
This page analyzes audience preferences based on content ratings. It displays the distribution of titles across different rating categories and helps identify the audience segments that Netflix primarily targets.

Netflix Global Content Analysis and Visualization using Python (Pandas) and Power BI

In this page, the first visualization created was a Donut Chart to represent audience preferences through the distribution of content ratings. For this visual, the rating column from the cleaned dataset (Netflix_df) was placed in the Legend section, while the Total Titles measure was added to the Values section. This enabled the chart to display the proportion of Netflix content across different audience rating categories.

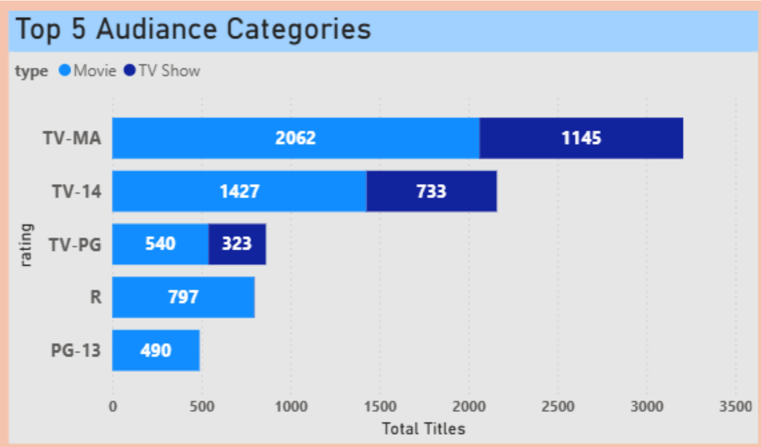


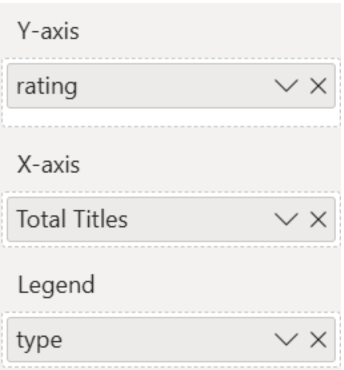
To improve readability and interpretability, the legend was positioned appropriately, and the data labels were configured to display both the title count and percentage contribution of each rating category. This visualization provides a clear understanding of the audience segments targeted by Netflix and highlights the relative share of content available for each rating classification.



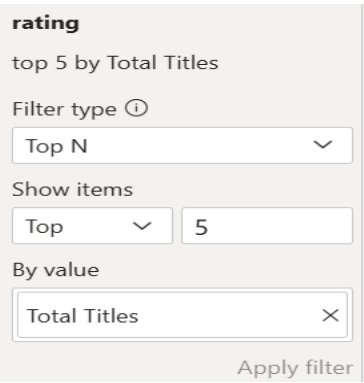
As the second visualization on this page, a Stacked Bar Chart was created to analyse the Top 5 Audience Rating Categories by Content Count and Content Type. In this visual, the Total Titles measure was placed on the X-axis, while the rating column from the cleaned dataset (Netflix_df) was placed on the Y-axis. The type column was added to the Legend section to differentiate the content into Movies and TV Shows, enabling a comparative analysis of content types across different audience rating categories.

Netflix Global Content Analysis and Visualization using Python (Pandas) and Power BI



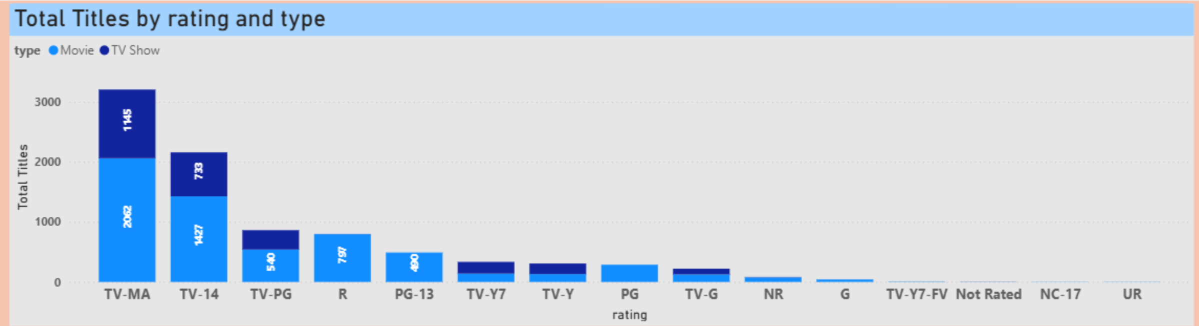


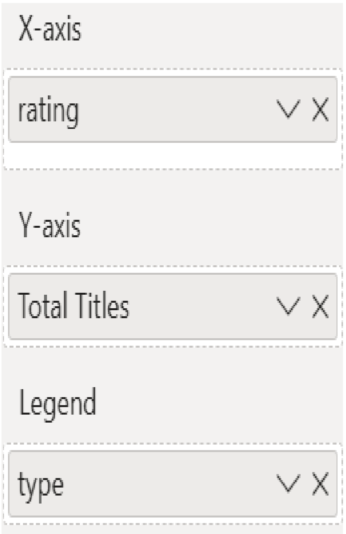
To enhance readability and accuracy, data labels were enabled and formatted as whole numbers. Furthermore, a Top N filter was applied to the rating column, where the value of N was set to 5, and the ranking was based on the unique count of show_id. This ensured that only the top five rating categories with the highest number of titles were displayed. The visualization provides a clear comparison of audience preferences and illustrates how Movies and TV Shows are distributed across the most prominent rating categories.

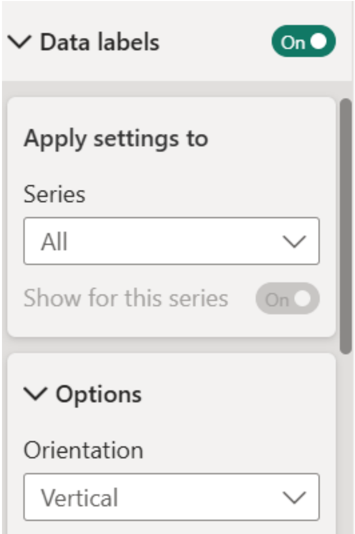


As the third visualization on this page, a Stacked Column Chart was created to illustrate the distribution of Netflix titles across different audience rating categories and content types. In this visual, the rating column from the cleaned dataset (Netflix_df) was placed on the X-axis, while the Total Titles measure was placed on the Y-axis. The type column was added to the Legend section to distinguish between Movies and TV Shows, enabling a comparative analysis of content distribution within each rating category.

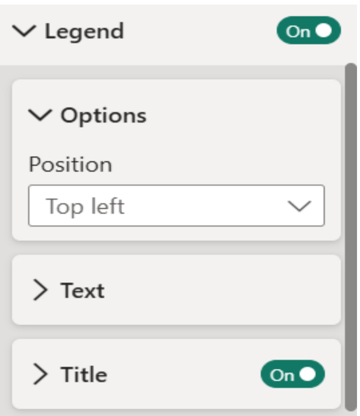
Netflix Global Content Analysis and Visualization using Python (Pandas) and Power BI







To improve the visual presentation, the legend was positioned appropriately for optimal readability, and data labels were enabled. The displayed values were formatted as whole numbers to ensure accurate and easy interpretation of the results. This visualization provides a comprehensive view of audience-targeted content and highlights the contribution of Movies and TV Shows across various rating classifications.

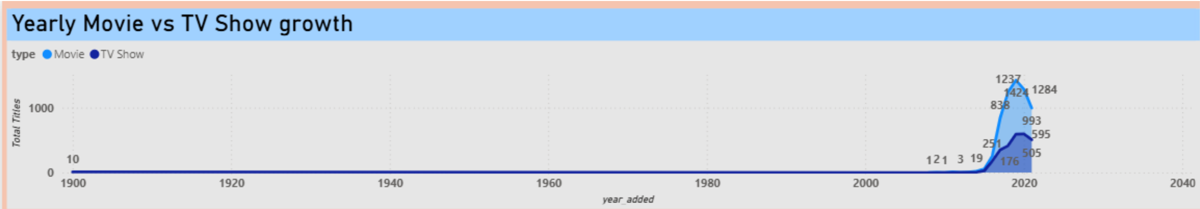


Page 5: Content Growth Overview

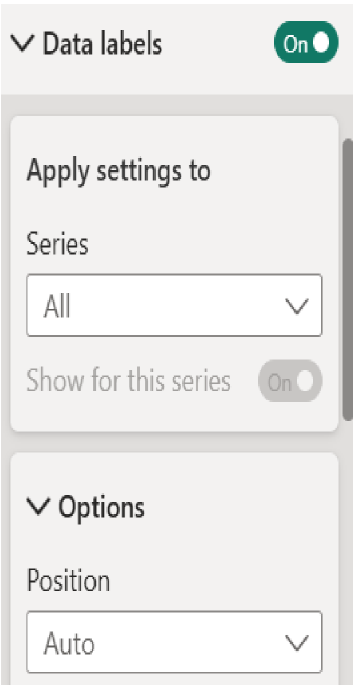
This page focuses on Netflix's content expansion over time. It presents year-wise content additions and compares the annual growth of Movies and TV Shows, providing insights into Netflix's content acquisition and expansion strategy.

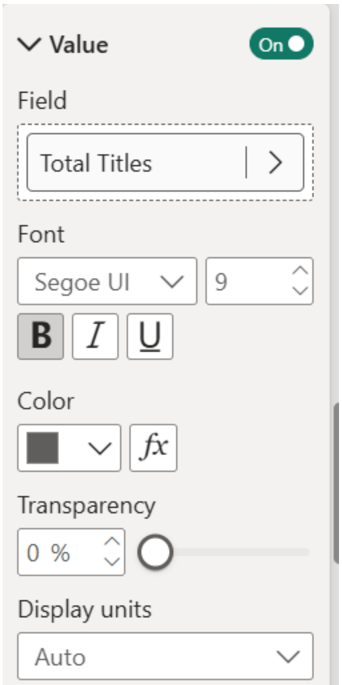
Netflix Global Content Analysis and Visualization using Python (Pandas) and Power BI

For this page, the first visualization created was a Line Chart to illustrate the yearly trend of content additions on Netflix and compare the growth of Movies and TV Shows over time. In this visual, the year_added column from the Netflix_df table was placed on the X-axis, while the Total Titles measure was placed on the Y-axis. The type column was added to the Legend section to distinguish between Movies and TV Shows, enabling a clear comparison of their annual growth trends.



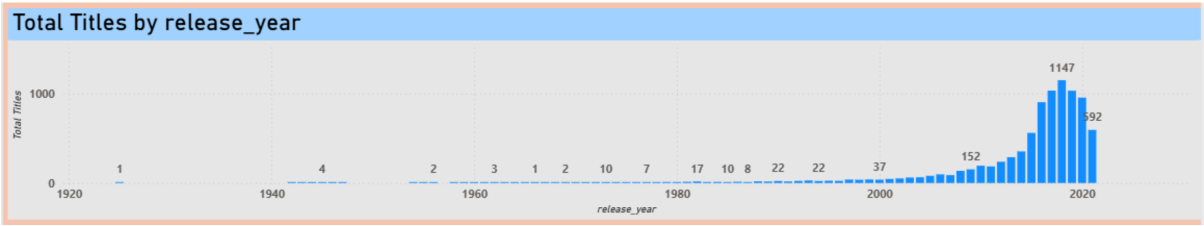
To improve the presentation and readability of the chart, titles were added and formatted for both the X-axis and Y-axis. Data labels were enabled to display the exact number of titles added each year, and the values were formatted as whole numbers to ensure accurate and easy interpretation. This visualization effectively highlights Netflix's content expansion strategy and the contribution of each content type to the platform's overall growth.



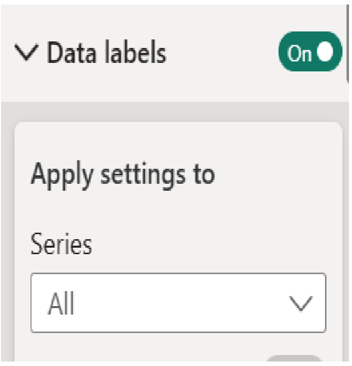


As the second visualization on this page, a Stacked Column Chart was created to display the distribution of Netflix titles by Release Year. In this visual, the release_year column from the Netflix_df table was placed on the X-axis, while the Total Titles measure was placed on the Y-axis. This enabled a year-wise analysis of the content available in the Netflix catalogue.

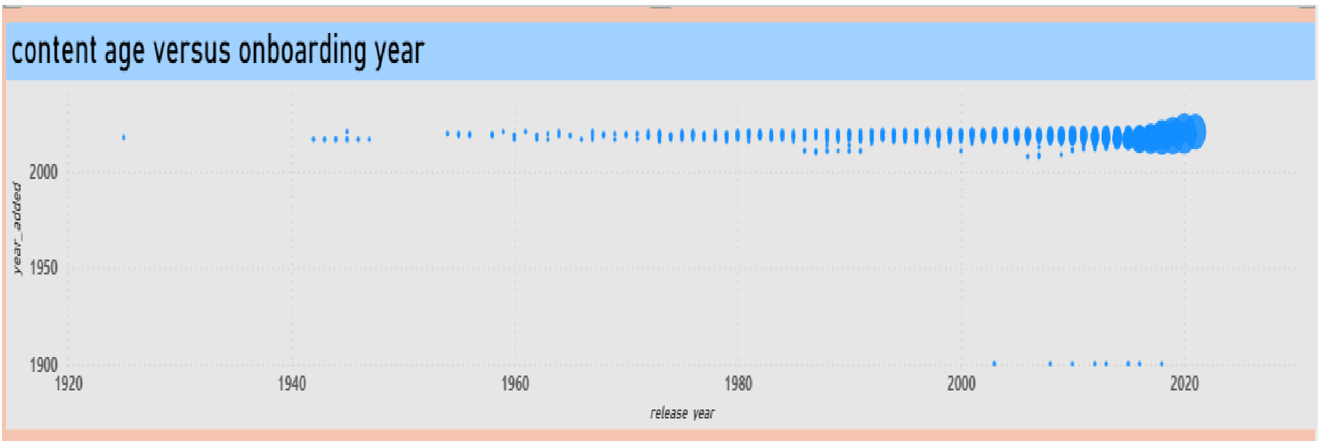
Netflix Global Content Analysis and Visualization using Python (Pandas) and Power BI



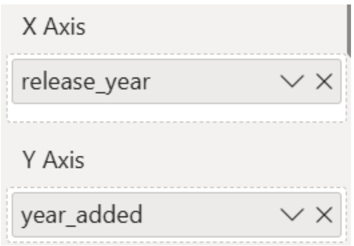
To improve readability and accuracy, data labels were enabled and formatted as whole numbers. The chart provides a clear view of content distribution across different release years, allowing users to identify trends, peak production periods, and the overall growth pattern of titles available on Netflix over time.



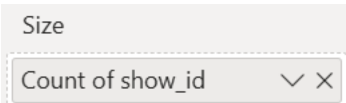
As the third visualization on this page, a Scatter Plot was created to analyse the relationship between a title's Release Year and the Year Added to Netflix. In this visual, the release_year column was placed on the X-axis, while the year_added column was placed on the Y-axis. This allowed for the comparison of when content was originally released versus when it became available on the Netflix platform.



Netflix Global Content Analysis and Visualization using Python (Pandas) and Power BI

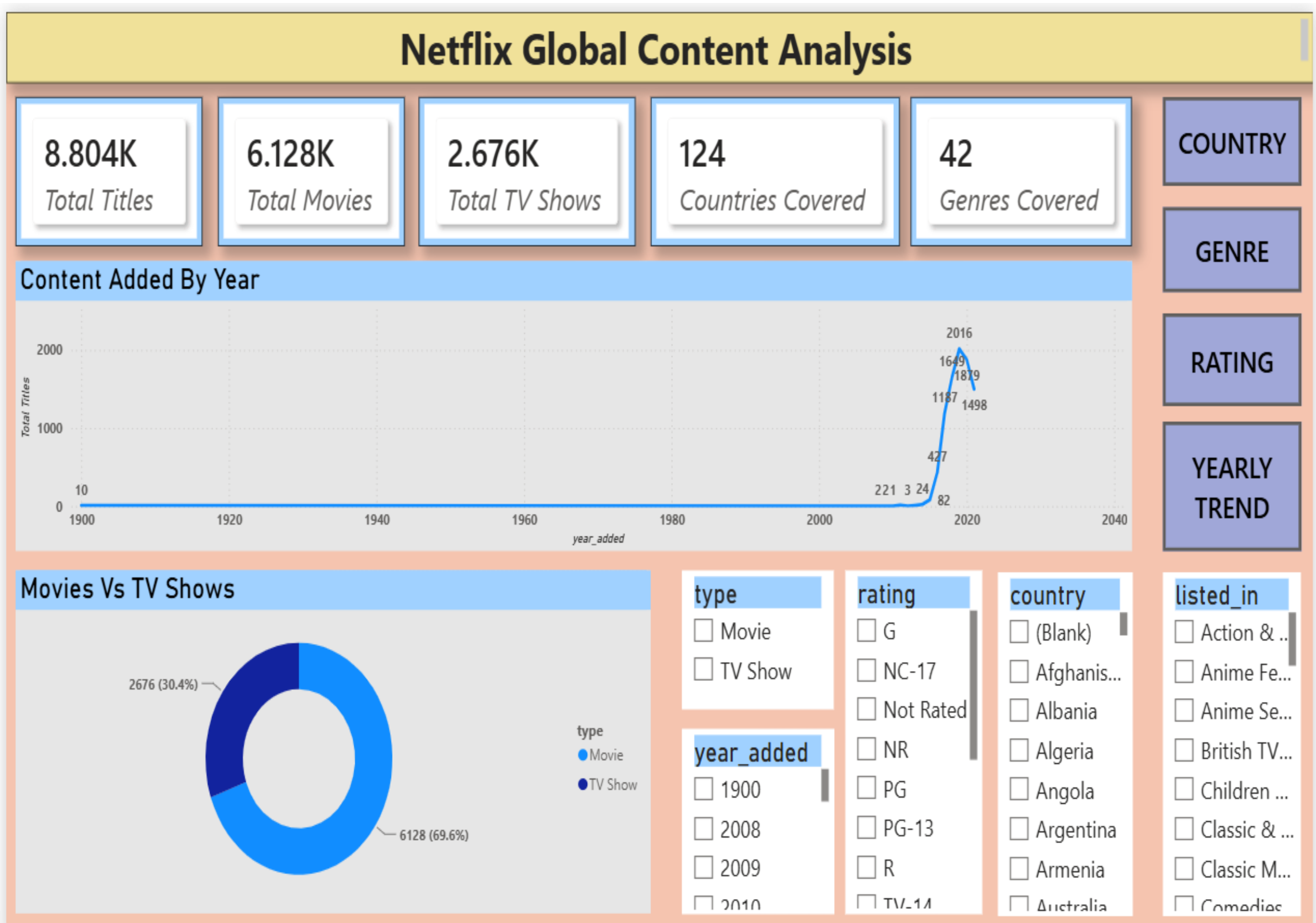


To provide additional analytical depth, the unique count of show_id was added to the **Size** section, causing the bubble size to vary based on the number of titles represented. Larger bubbles indicate a higher concentration of titles for a particular combination of release year and year added. This visualization helps identify whether Netflix tends to add content shortly after its release or after a significant delay, while also highlighting the volume of titles associated with these trends through bubble size variations.



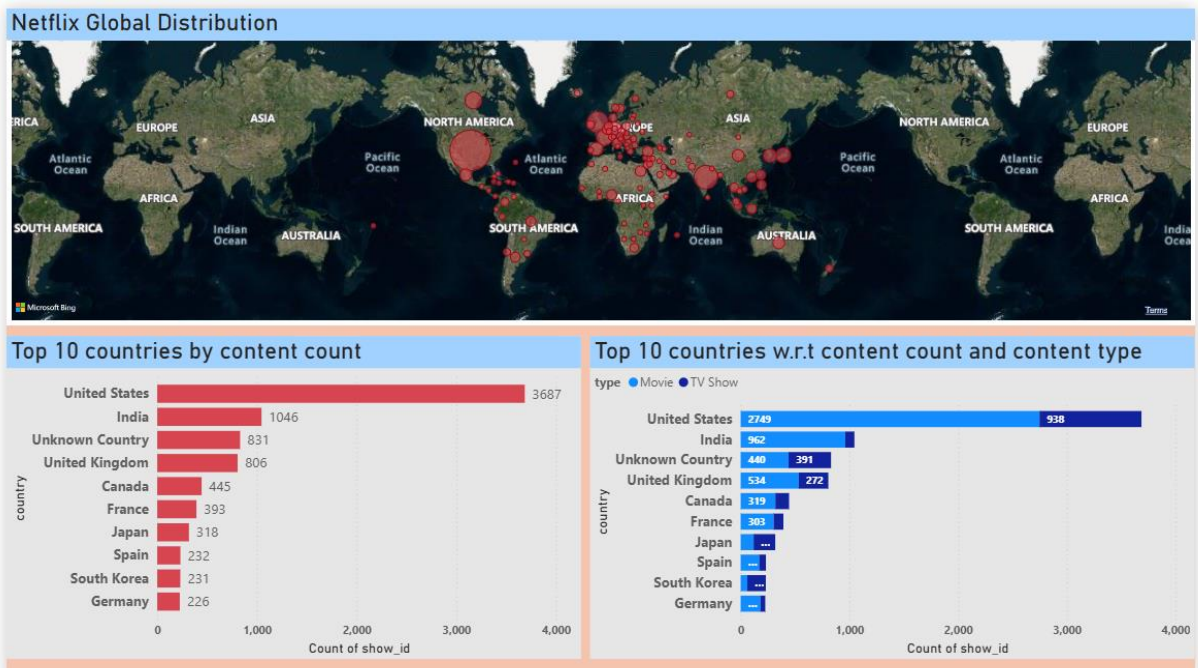
The five dashboard pages developed as part of this project are presented below.

Netflix Overview Page:

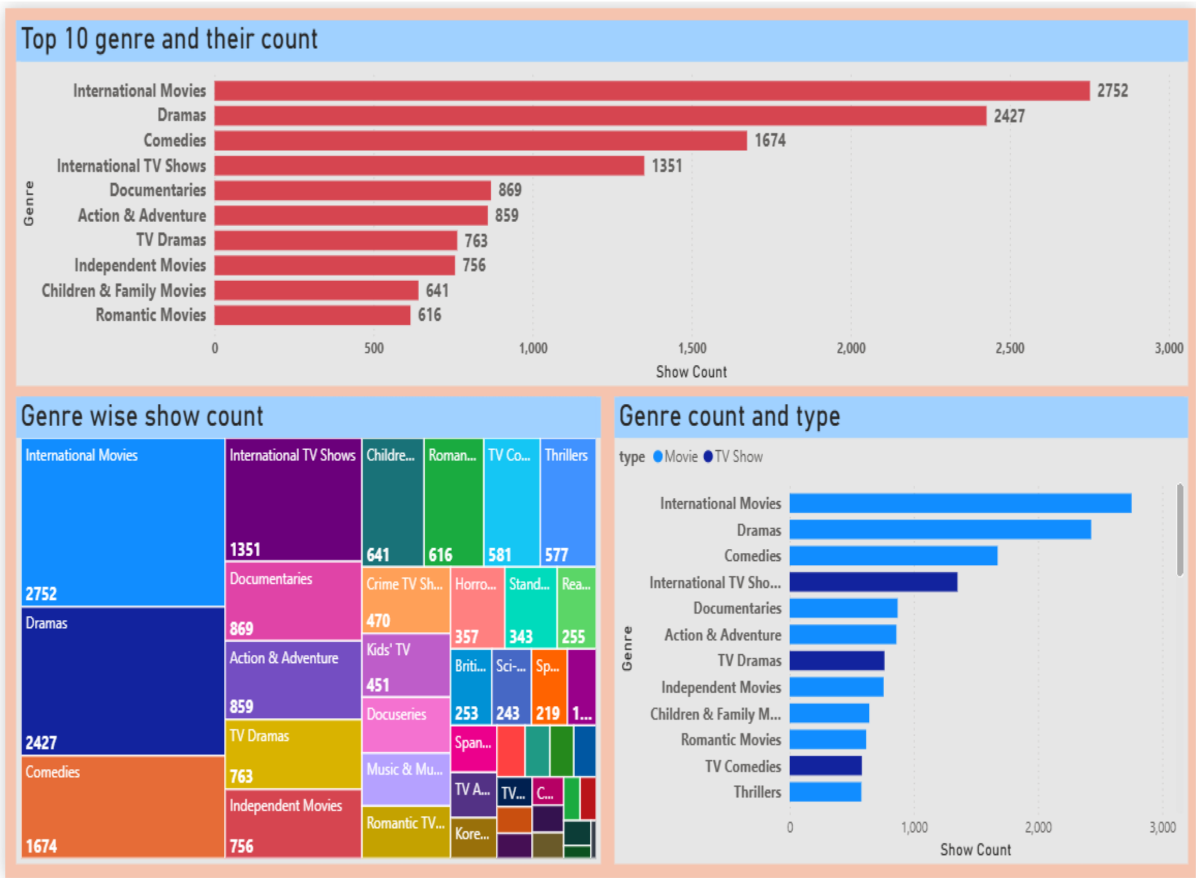


Netflix Global Content Analysis and Visualization using Python (Pandas) and Power BI

Country Overview Page:

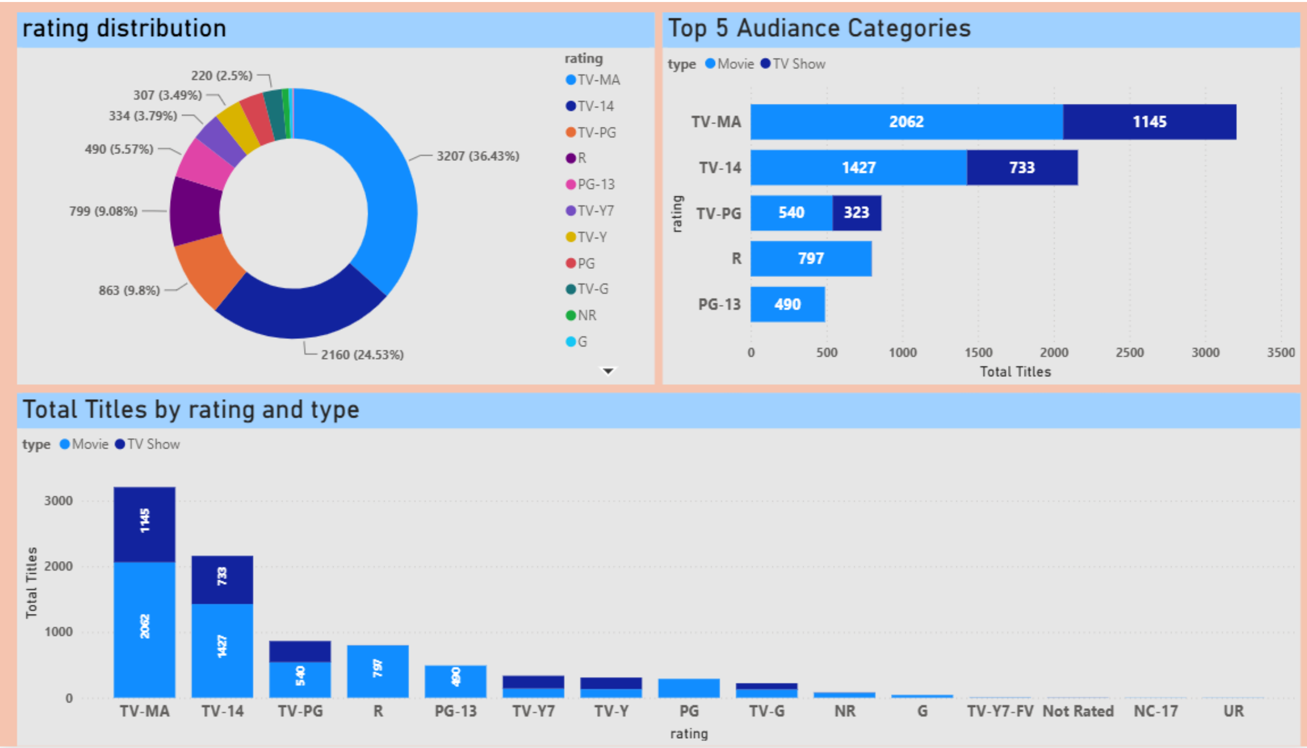


Genre Overview Page:

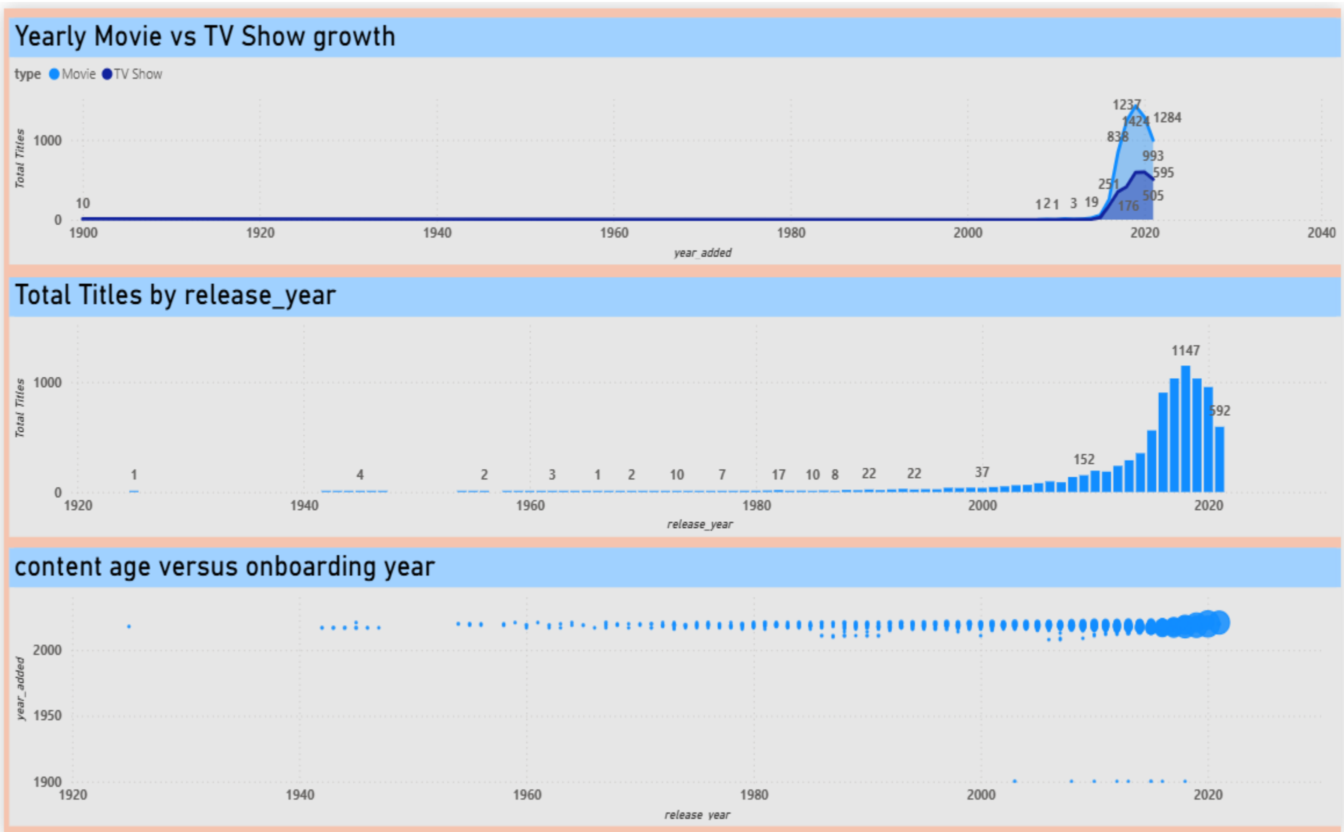


Netflix Global Content Analysis and Visualization using Python (Pandas) and Power BI

Audience Rating Overview Page:



Yearly Trend Page:



Netflix Global Content Analysis and Visualization using Python (Pandas) and Power BI

9. Recommendations and Future Scope:

1. Global Diversification

The analysis indicates that Netflix has successfully expanded its content portfolio across multiple countries, demonstrating strong global diversification. To further strengthen its market presence, Netflix should continue investing in localized and region-specific content that aligns with the cultural preferences of emerging markets. In the future, the analysis can be extended by incorporating country-wise viewer engagement, subscription growth, and retention metrics to better understand the effectiveness of regional content strategies.

2. Movie-Heavy Catalogue

The findings reveal that Netflix's content library is dominated by movies compared to TV shows. To improve long-term audience retention and engagement, Netflix should increase the production and acquisition of high-quality TV series, as serialized content generally encourages repeat viewership. Future research can focus on analysing watch-time, completion rates, and popularity metrics to determine the optimal balance between movies and TV shows within the platform.

3. Growing International Focus

The increasing contribution of international content highlights Netflix's strategic shift toward global audiences. To capitalize on this trend, Netflix should prioritize multilingual content production and strengthen partnerships with international studios and production houses. Future studies can examine regional content performance, cultural preferences, and audience behaviour patterns to identify opportunities for further international expansion.

4. Strong Mature Audience Segment

The rating analysis suggests that a significant portion of Netflix's content is targeted toward mature audiences. While this strategy caters to a large viewer segment, Netflix can broaden its reach by expanding its collection of family-friendly, children, and teen-oriented content. Future analysis can include audience segmentation based on age groups, viewing habits, and content preferences to support more targeted content development strategies.

5. Rapid Content Growth

The year-wise content analysis demonstrates substantial growth in Netflix's content additions over time. To maintain sustainable growth, Netflix should focus on balancing content quantity with quality and relevance to audience preferences. Future work can leverage predictive analytics and demand forecasting techniques to anticipate content trends, optimize investment decisions, and improve content acquisition planning.

Netflix Global Content Analysis and Visualization using Python (Pandas) and Power BI